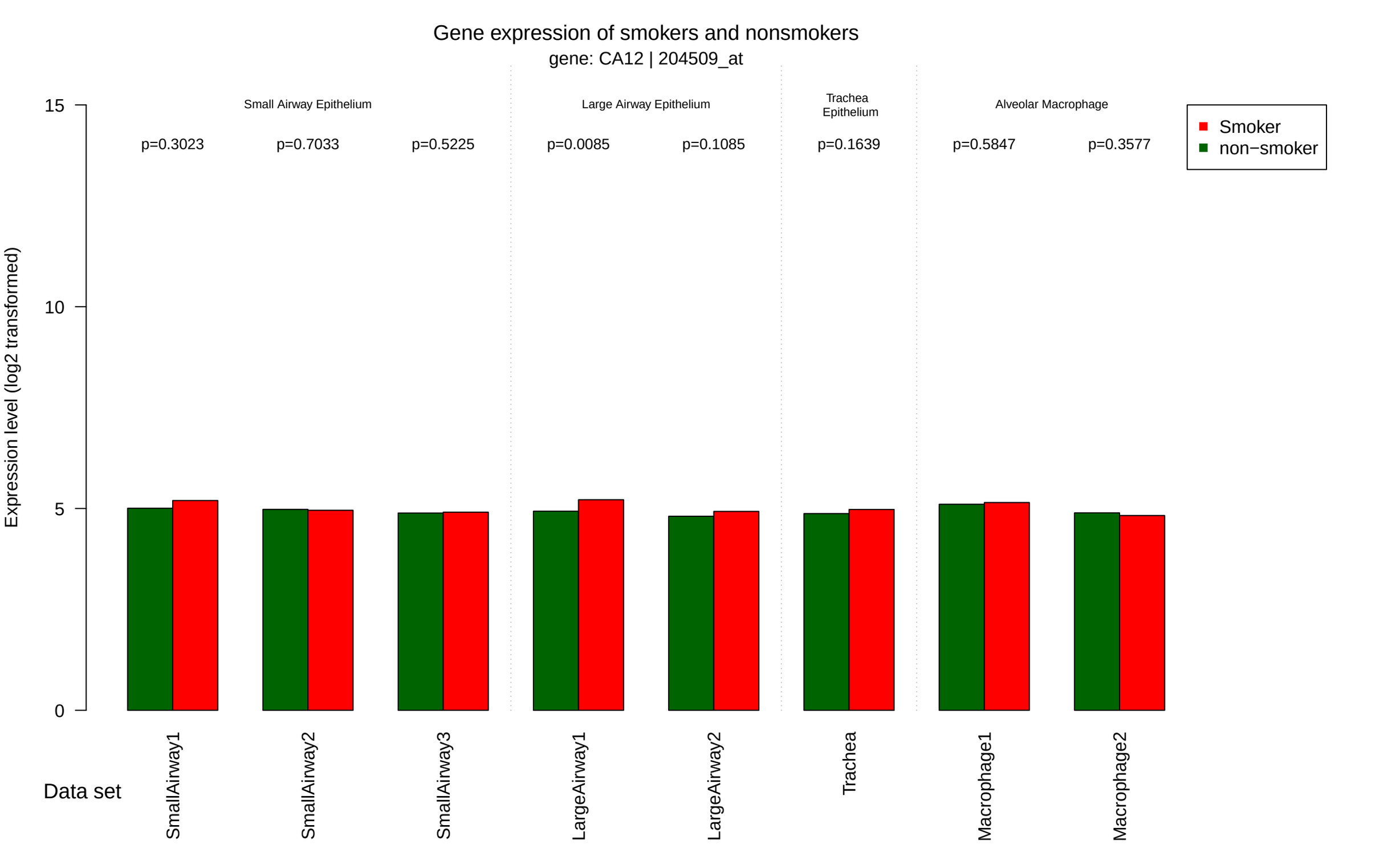
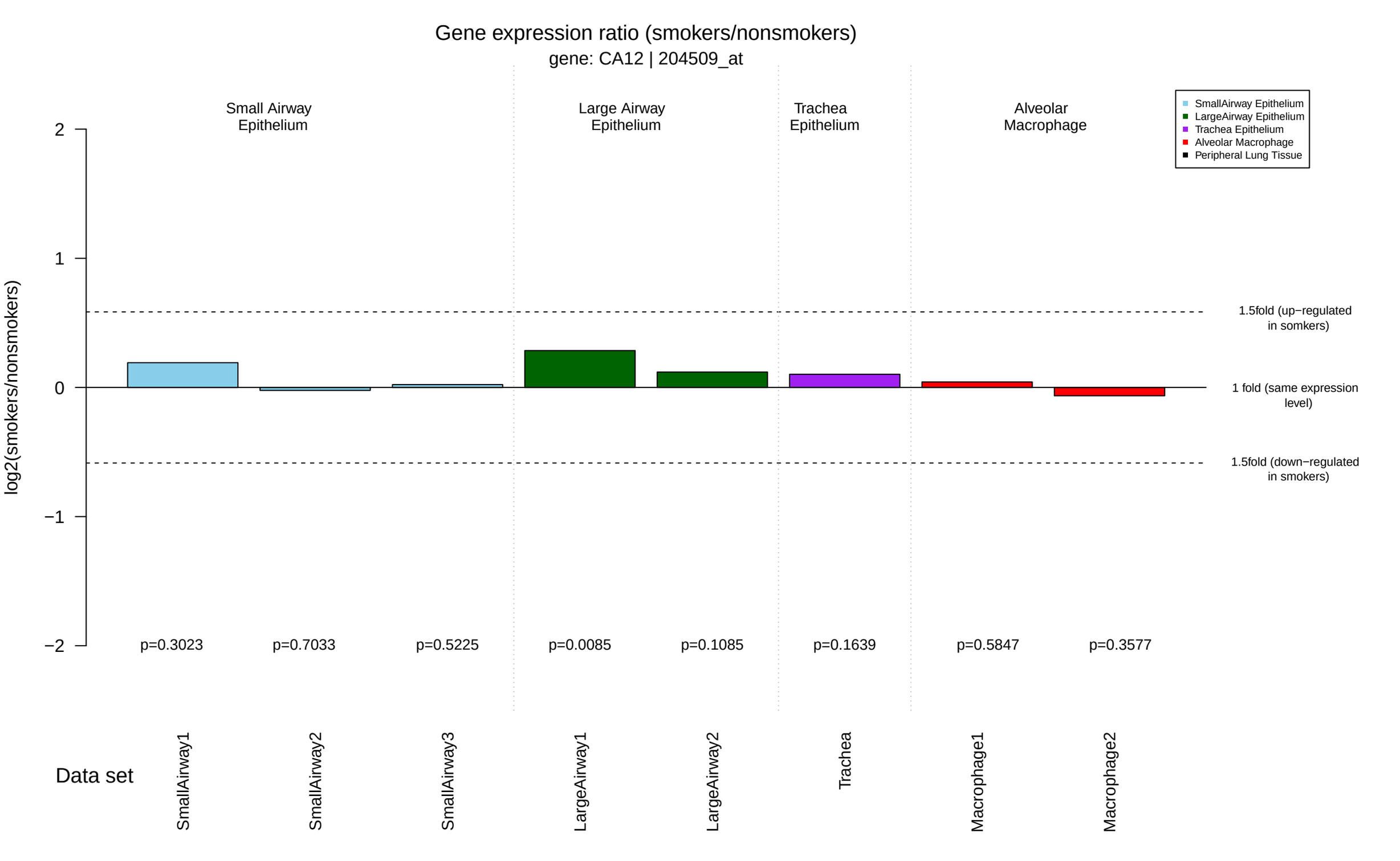
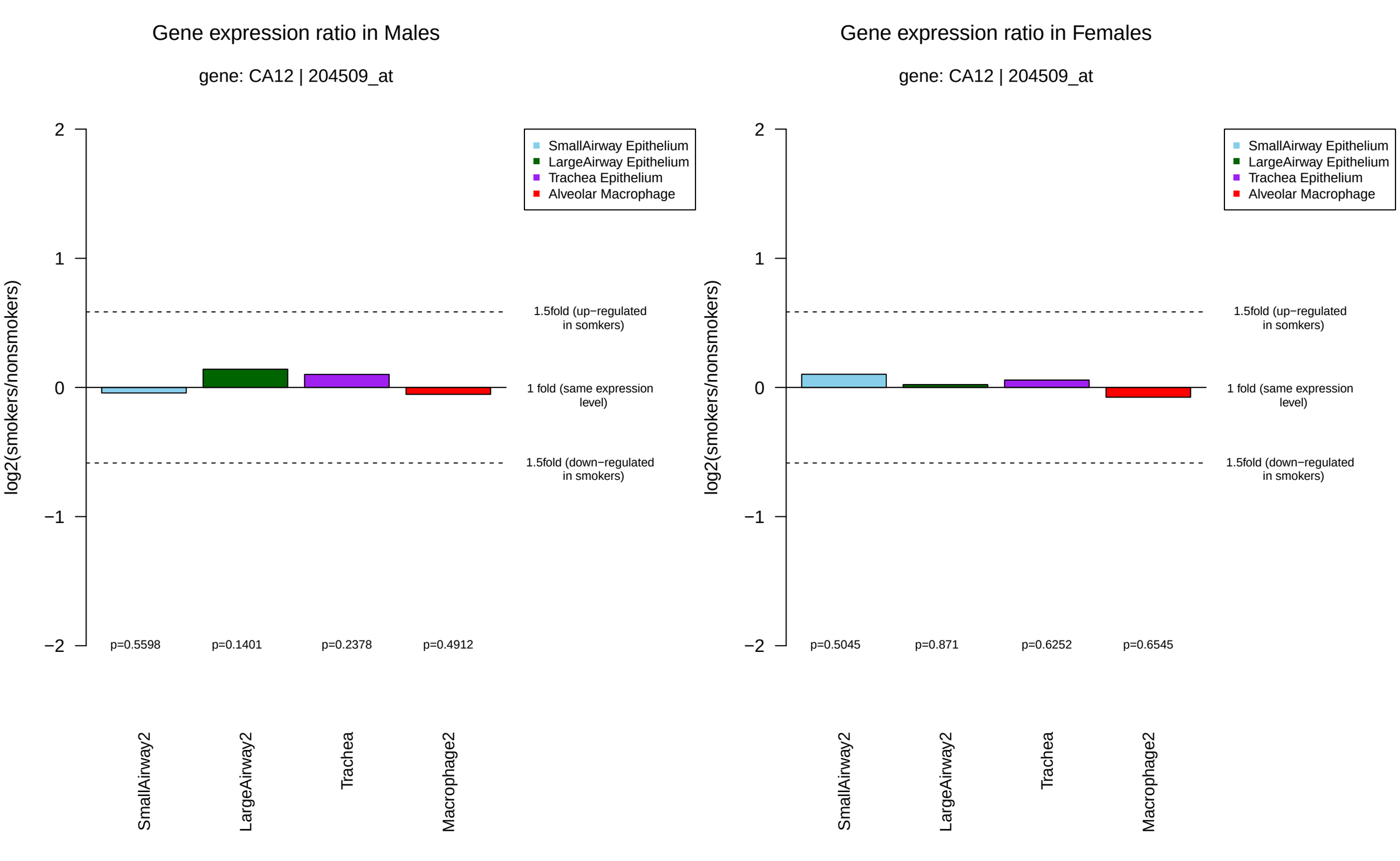
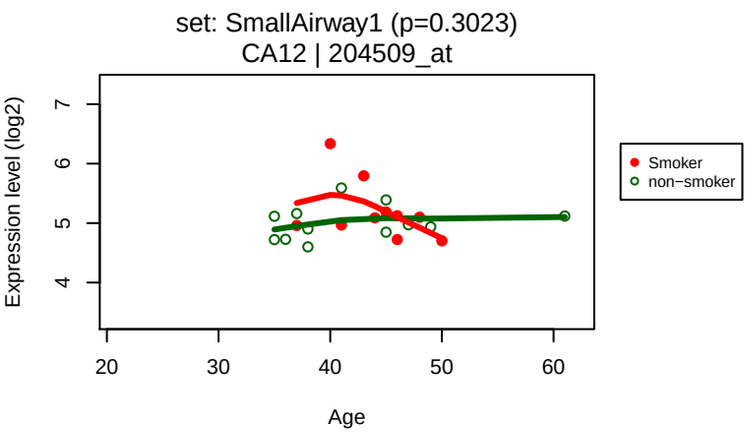
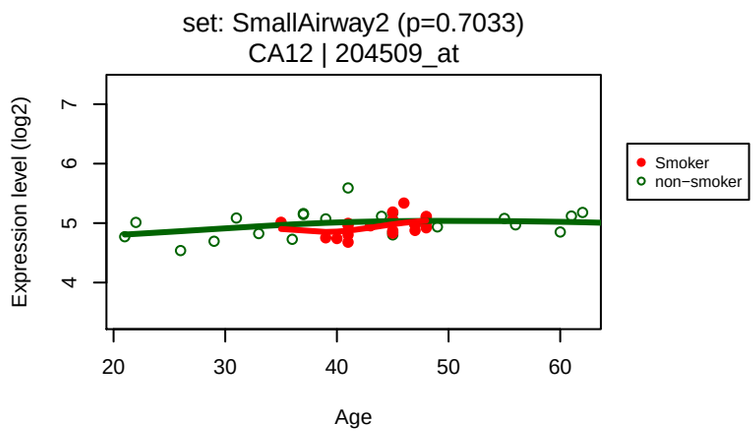
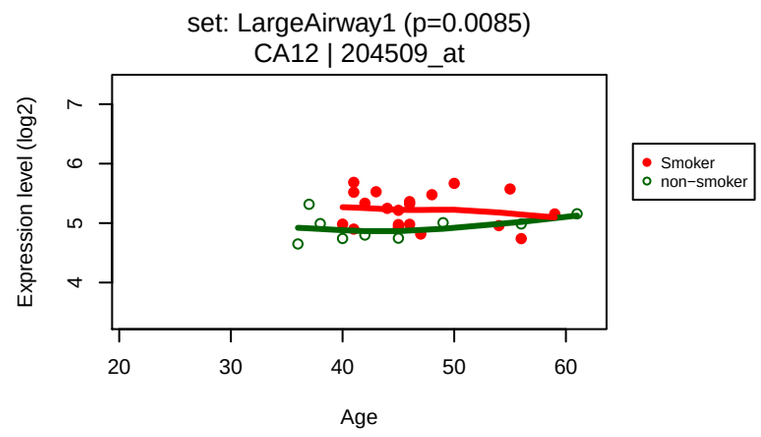

Gene expression of smokers and nonsmokers
gene: CA12 | 204509_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3023
p=0.7033
p=0.5225
p=0.0085
p=0.1085
p=0.1639
p=0.5847
p=0.3577
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

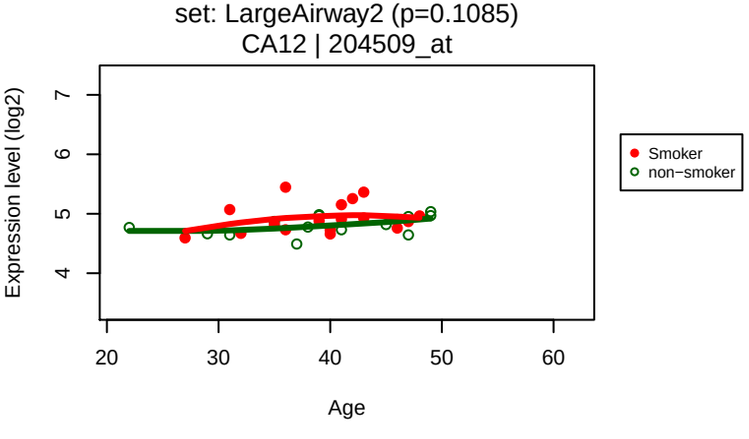

Gene expression ratio (smokers/nonsmokers)
gene: CA12 | 204509_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3023
p=0.7033
p=0.5225
p=0.0085
p=0.1085
p=0.1639
p=0.5847
p=0.3577
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

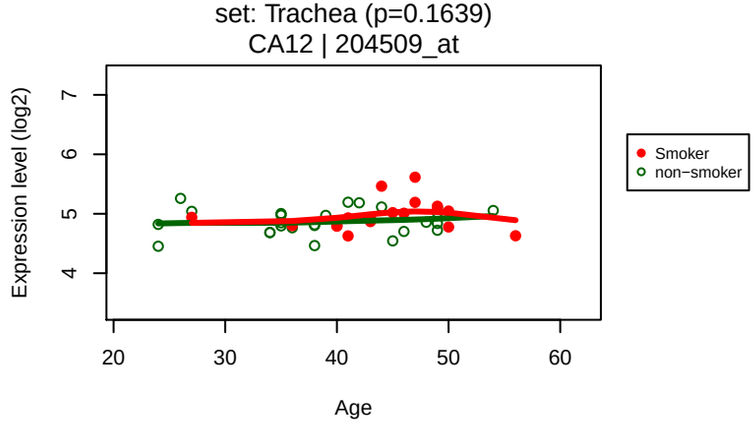

Gene expression ratio in Males
gene: CA12 | 204509_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5598
p=0.1401
p=0.2378
p=0.4912
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CA12 | 204509_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5045
p=0.871
p=0.6252
p=0.6545
SmallAirway2
LargeAirway2
Trachea
Macrophage2

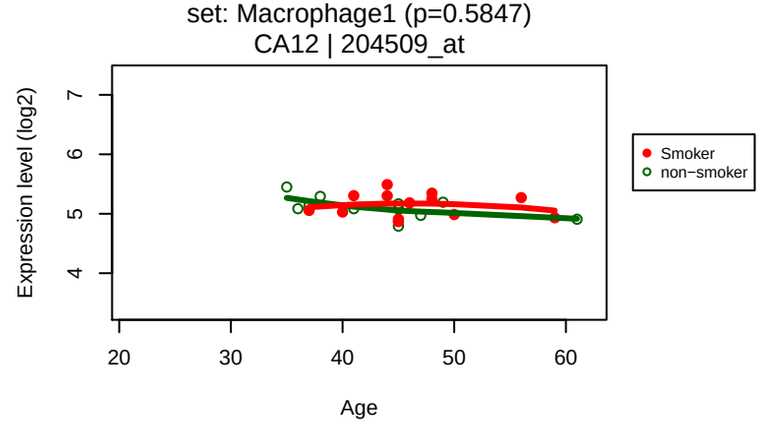

set: SmallAirway1 (p=0.3023)
CA12 | 204509_at
Expression level (log2)
Age
Smoker
non−smoker

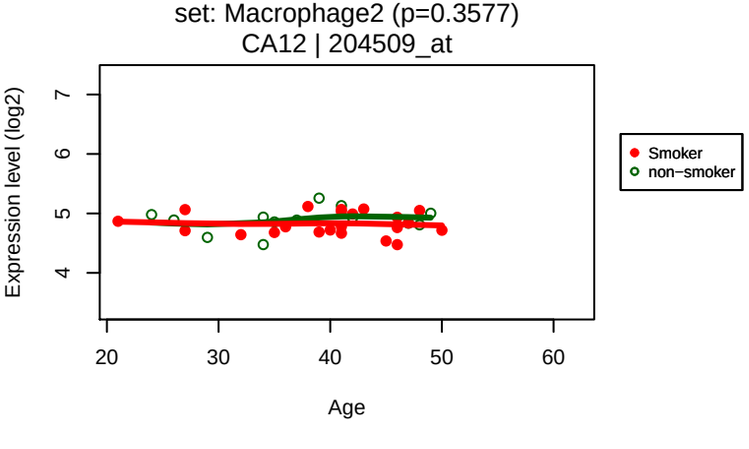

set: SmallAirway2 (p=0.7033)
CA12 | 204509_at
Expression level (log2)
Age
Smoker
non−smoker

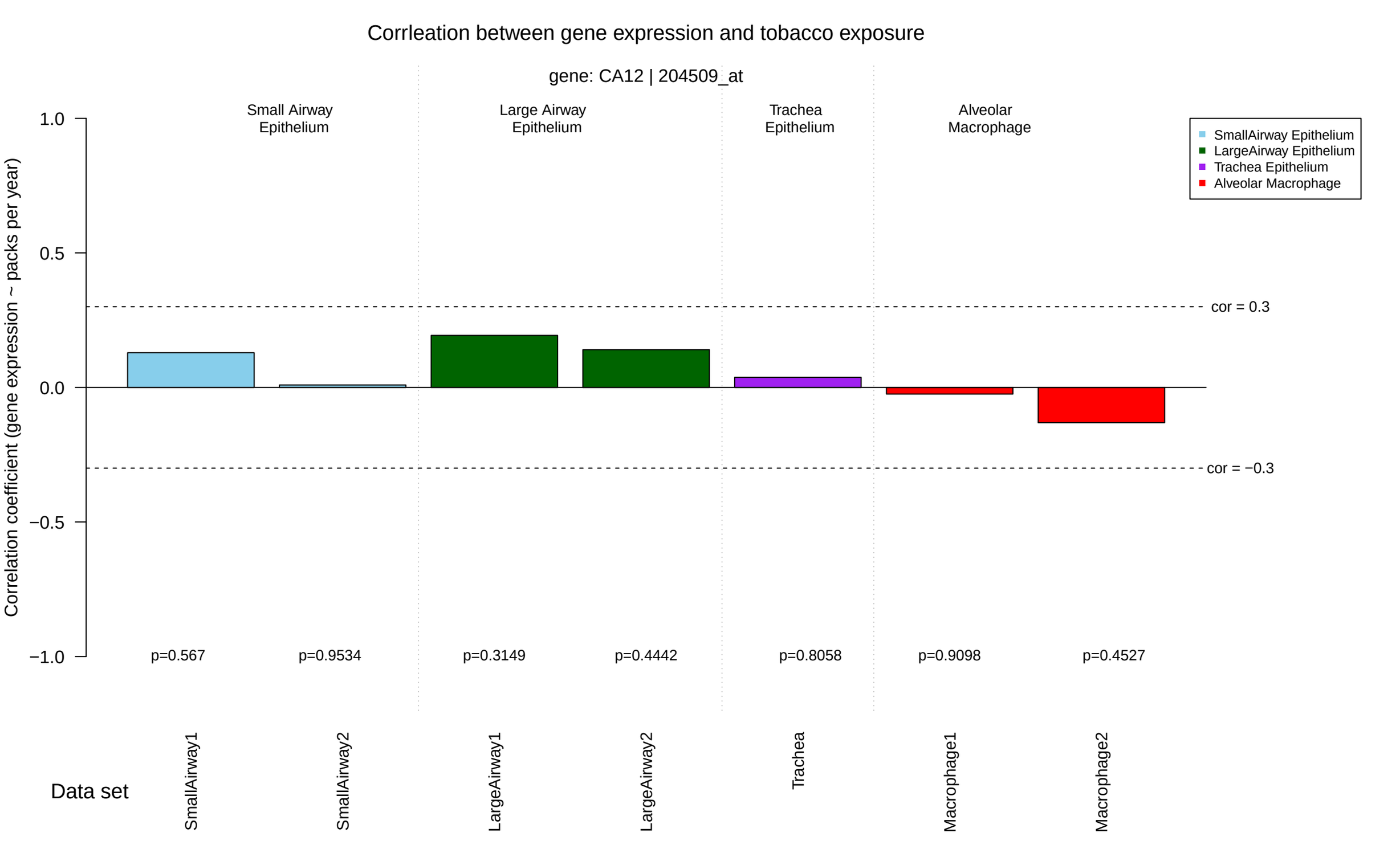

set: LargeAirway1 (p=0.0085)
CA12 | 204509_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1085)
CA12 | 204509_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1639)
CA12 | 204509_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5847)
CA12 | 204509_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3577)
CA12 | 204509_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CA12 | 204509_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.567
p=0.9534
p=0.3149
p=0.4442
p=0.8058
p=0.9098
p=0.4527
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2